Yingrui Hou

Data Science & Scientific Data Engineering

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience building Python/C++ analysis workflows for noisy experimental data. Combines statistical inference, machine learning, data preparation and model validation to produce reproducible results. Experienced in adapting methods across research problems, coordinating connected analysis tasks and explaining findings in international teams. Targeting data science and engineering roles where reliable analysis, reusable processing and careful validation matter.

Core skills

- Data & programming: Python, NumPy, pandas, scikit-learn, statsmodels, matplotlib/seaborn; C/C++, ROOT/RooFit.
- Modelling: likelihood and mixture models, statistical inference, uncertainty analysis, CatBoost/gradient boosting, feature selection and distribution reweighting.
- Workflow engineering: reusable processing components, configuration-driven analysis, Shell/YAML/CMake, Linux, shared-code review and reproducible model application.
- Validation: train/test diagnostics, threshold studies, control-sample checks, calibration, bias correction and root-cause investigation.

Research experience

Postdoctoral Researcher | 2021–2025

Research in CNRS/CERN environments; LHCb and GRAiNITA collaborations.

- Developed reusable Python/C++/ROOT workflows linking data preparation, weak-signal extraction, likelihood fitting and validation when event-level ground truth was unavailable.
- Built CatBoost selection workflows with feature checks, train/test comparisons and threshold scans; saved models and configurations for reproducible application.
- Aligned simulated, control and target datasets through gradient-boosting reweighting and calibration; propagated residual modelling differences into uncertainty assessments.
- Investigated reconstruction discrepancies using benchmark channels and data/simulation comparisons; identified detector and software issues that informed corrective follow-up.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

Machine learning under dataset shift

Problem: training and calibration samples did not naturally represent the target events. Implemented CatBoost classifiers, score checks and sample reweighting as a connected workflow. Train/test response comparisons checked overtraining before threshold selection; alignment supported subsequent inference.

Bias-aware inference from imperfect measurements

Implemented likelihood components for selection efficiency and measurement resolution, using separate corrections for Run 1 and Run 2. Control-sample alignment, joint fits and uncertainty checks made assumptions explicit and reusable across datasets. This experience transfers to statistical modelling and model-risk assessment.

Measured data transformed into model inputs

Built a beam-test measurement-to-simulation chain: derived position-dependent response maps and integrated them into Geant4/C++ analysis. Quantified a non-uniformity contribution below 1% to the energy-resolution constant term in the first GRAiNITA prototype study. The result belongs to that prototype and collaborative study.

Research training applied to engineering

- Adaptability and learning: moved between likelihood inference, CatBoost-based selection and Geant4 detector modelling; combined literature, implementation and data checks to apply methods to unfamiliar problems.
- Research project coordination: organised connected analysis tasks from measurement inputs to validated results, with reusable components and documented assumptions for collaboration review.
- Presentation and international teamwork: presented at GDR-InF 2022 and the 5th FCC/DRD France Workshop in 2025, including a talk on behalf of GRAiNITA; contributed to shared code and technical reviews.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- Machine learning under dataset shift (https://yingrui-hou.github.io/PersonalPage/index.html#work-3)
- Bias-aware inference from imperfect measurements (https://yingrui-hou.github.io/PersonalPage/index.html#work-2)
- Measured data transformed into model inputs (https://yingrui-hou.github.io/PersonalPage/index.html#work-4)